

Theorem (The Fundamental Theorem of Calculus, Part 1 (FTC1))

Statement:

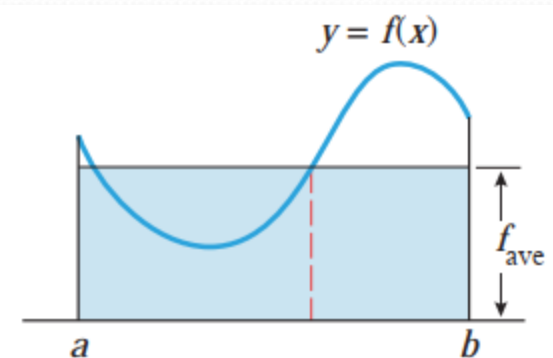
If f is continuous on $[a, b]$ and F is any *anti-derivative* of f on $[a, b]$, then

$$\int_a^b f(x)dx = F(b) - F(a) \quad (1)$$

Average Value:

If f is continuous on $[a, b]$, then the **average value** (or **mean value**) of f on $[a, b]$ is defined to be

$$f_{\text{ave}} = \frac{1}{b-a} \int_a^b f(x)dx$$



Problem: Compute the average value of $f(x) = \sin(x)$ on the interval $[0, \pi]$.

Solution:

Given, $f(x) = \sin(x)$, $a = 0$, $b = \pi$

we know,

$$\begin{aligned} f_{ave} &= \frac{1}{b-a} \int_a^b f(x) dx \\ &= \frac{1}{\pi-0} \int_0^{\pi} \sin(x) dx \\ &= \frac{1}{\pi} \left[-\cos(x) \right]_0^{\pi} \\ &\approx 0.6366198 \end{aligned}$$

Problem: In the Dhaka city, the temperature (in $^{\circ}\text{C}$) recorded during a day followed the curve

$$T = 0.001t^4 - 0.28t^2 + 25$$

, where where t is the number of hours from noon ($0 \leq t \leq 12$). What was the average temperature during the day?

Solution:

Given, $T = 0.001t^4 - 0.28t^2 + 25$,

$a = 0$, $b = 12$

we know,

$$\begin{aligned} f_{ave} &= \frac{1}{b-a} \int_a^b f(x) dx \\ &= \frac{1}{12-0} \int_0^{12} (0.001t^4 - 0.28t^2 + 25) dx \\ &= \frac{1}{12} \left[\frac{0.001t^5}{5} - \frac{0.28t^3}{3} + 25t \right]_0^{12} \\ &\approx 15.7^{\circ}\text{C} \end{aligned}$$

Problem: Applying the FTC1 to estimate the following integrals:

$$(i) \int_0^2 (x^2 - 2x) dx, (ii) \int_{-1}^4 f(x) dx; f(x) = \begin{cases} x + 1, & -1 \leq x < 0 \\ x, & 0 \leq x < 2 \\ 3, & 2 \leq x \leq 4 \end{cases}, (iii) \int_0^3 f(x) dx; f(x) = |x - 2|$$

Solution:

(i) The function $f(x) = x^2 - 2x$ is continuous on $[0, 2]$, so we get

$$\begin{aligned} \int_0^2 (x^2 - 2x) dx &= \left(\frac{x^3}{3} - x^2 \right) \Big|_0^2 \\ &= \left(\frac{8}{3} - 4 \right) - 0 \\ &= -\frac{4}{3} \end{aligned}$$

(ii) The function $f(x)$ is piecewise continuous on $[-1, 4]$, so we get

$$\begin{aligned} \int_{-1}^4 f(x) dx &= \int_{-1}^0 f(x) dx + \int_0^2 f(x) dx + \int_2^4 f(x) dx \\ &= \int_{-1}^0 (x + 1) dx + \int_0^2 x dx + \int_2^4 3 dx \\ &= \left(\frac{x^2}{2} + x \right) \Big|_{-1}^0 + \left(\frac{x^2}{2} \right) \Big|_0^2 + (3x) \Big|_2^4 \\ &= \frac{17}{2} \end{aligned}$$

Solution:

(iii) We've,

$$\begin{aligned} f(x) = |x - 2| &= \begin{cases} (x - 2), & x - 2 \geq 0 \\ -(x - 2), & x - 2 < 0 \end{cases} \\ &= \begin{cases} x - 2, & x \geq 2 \\ -x + 2, & x < 2 \end{cases} \end{aligned}$$

Since the function $f(x)$ is continuous on $[0, 3]$, so we get

$$\begin{aligned} \int_0^3 f(x) dx &= \int_0^3 |x - 2| dx \\ &= \int_0^2 |x - 2| dx + \int_2^3 |x - 2| dx \\ &= \int_0^2 (-x + 2) dx + \int_2^3 (x - 2) dx \\ &= \left(-\frac{x^2}{2} + 2x \right) \Big|_0^2 + \left(\frac{x^2}{2} - 2x \right) \Big|_2^3 \\ &= \frac{5}{2} \end{aligned}$$

Real-life Applications:

Problem: Breathing is cyclic and a full respiratory cycle from the beginning of inhalation to the end of exhalation takes about 5 s. The maximum rate of air flow into the lungs is about 0.6 L/s. Suppose for a man, the rate of air flow into the lungs is modeled by the function

$$f(t) = \frac{1}{2} \sin\left(\frac{2\pi t}{5}\right)$$

Apply this model to find the volume of inhaled air in the lungs at time t , and compute the average volume of inhaled air in the lungs in one respiratory cycle.

Solution: Given,

$$f(t) = \frac{1}{2} \sin\left(\frac{2\pi t}{5}\right)$$

The volume of inhaled air in the lungs at time t is

$$\begin{aligned} V(t) &= \int_0^t f(u) du \\ &= \int_0^t \frac{1}{2} \sin\left(\frac{2\pi u}{5}\right) du \end{aligned}$$

$$V(t) = \int_0^t f(u) du$$

$$= \int_0^t \frac{1}{2} \sin\left(\frac{2\pi u}{5}\right) du$$

$$= \int_0^{\frac{2\pi t}{5}} \frac{1}{2} \sin(v) \cdot \frac{5}{2\pi} dv$$

$$= \frac{5}{4\pi} \int_0^{\frac{2\pi t}{5}} \sin(v) dv$$

$$= \frac{5}{4\pi} (-\cos(v)) \Big|_0^{\frac{2\pi t}{5}}$$

$$= \frac{5}{4\pi} \left(1 - \cos\left(\frac{2\pi t}{5}\right) \right)$$

Let,

$$v = \frac{2\pi u}{5}$$

$$\text{or, } \frac{dv}{du} = \frac{2\pi}{5}$$

$$\text{or, } du = \frac{5}{2\pi} dv$$

$$u = 0$$

$$v = 0$$

$$u = t$$

$$v = \frac{2\pi t}{5}$$

Average:

$$V_{ave} = \frac{1}{5-0} \int_0^5 V(t) dt$$

$$= \frac{1}{5} \int_0^5 \frac{5}{4\pi} \left(1 - \cos\left(\frac{2\pi t}{5}\right) \right) dt$$

$$= \frac{1}{4\pi} \left(t - \frac{5}{2\pi} \sin\left(\frac{2\pi t}{5}\right) \right) \Big|_0^5$$

$$= \frac{1}{4\pi} [(5-0) - 0]$$

$$= \frac{5}{4\pi}$$

$$\approx 0.4L$$

Problems: The concentration of a certain drug (in mg/cc) in a patient's bloodstream t hr after injection is

$$C(t) = \frac{0.2t}{t^2 + 1}$$

Determine the average concentration of the drug in the patient's bloodstream over the first 4 hr after the drug is injected.

Answer: approximately 0.071 mg/cc

Absorption of Drugs The concentration of a drug in an organ at any time t , in seconds) is given by

$$C(t) = \begin{cases} 0.3t - 18(1 - e^{-t/60}) & \text{if } 0 \leq t \leq 20 \\ 18e^{-t/60} - 12e^{-(t-20)/60} & \text{if } t > 20 \end{cases}$$

where $C(t)$ is measured in grams per cubic centimeter (g/cm^3). Find the average concentration of the drug in the organ over the first 30 sec after it is administered.

Daylight Hours in Chicago The number of hours of daylight at any time t in Chicago is approximated by

$$L(t) = 2.8 \sin\left[\frac{2\pi}{365}(t - 79)\right] + 12$$

where t is measured in days and $t = 0$ corresponds to January 1. What is the daily average number of hours of daylight in Chicago over the year? Over the summer months from June 21 ($t = 171$) through September 20 ($t = 262$)?

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Water Level in Boston Harbor The water level (in feet) in Boston Harbor during a certain 24-hr period is approximated by the formula

$$H = 4.8 \sin\left[\frac{\pi}{6}(t - 10)\right] + 7.6 \quad 0 \leq t \leq 24$$

where $t = 0$ corresponds to 12 A.M. What is the average water level in Boston Harbor over the 24-hr period on that day?

Canadian Oil-Sands Production The production of oil (in millions of barrels per day) extracted from oil sands in Canada is projected to be

$$P(t) = \frac{4.76}{1 + 4.11e^{-0.22t}} \quad 0 \leq t \leq 15$$

where t is measured in years, with $t = 0$ corresponding to the beginning of 2005. What will the total oil production of oil from oil sands be over the years from the beginning of 2005 until the beginning of 2020 ($t = 15$)?



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